

Cryptology Research Paper
Quantum Computers and Public Key Cryptography
EN.695.641.81.SP26
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Overspecialization leads to death. Evolutionarily, if one species becomes isolated from another via geography, be it mountains or forests; behaviorally, like different bird songs that are no longer recognized; or by physicality, such as coloration of a mating display that is no longer acceptable. When a new species comes into being, it fills a niche in the environment. The environment changes over geological timescales, which accrue slowly, and if a species doesn't adapt, it goes extinct.

Fairly recently, quantum computers have supposedly been used to factor an 80-bit RSA modulus¹ via Shor's algorithm, which would mean that before long, algorithms whose security relies on the difficulty of integer factorization can no longer be trusted as secure for encryption. Shor's 1994 paper² introduced two related quantum algorithms for the discrete logarithmic problem and integer factorization. It presents the discrete-logarithm result first, with both ultimately being instances of a more general period-finding subroutine that is the genuinely 'quantum' part of the algorithm. If an attacker factories n , then they can break RSA. However, it's not guaranteed that is the only way. Thus, the continual improvement in both the theoretical algorithms and quantum computing power is akin to the changing environment in the natural world, with the RSA family of algorithms slowly becoming unfit in a world where they can no longer be relied on as a deterrent to data encryption.

RSA³ relies on using two large prime numbers as factors during key generation, p and q . The specific values chosen depend on the specific algorithm and the requirement for security. Those prime factors when multiplied together form the modulus, n , for both the private and public keys. The modulus is not hidden and is part of the public key. The chosen prime values p and q then each have their value, minus

¹ Quantum Computing - How it Changes Encryption as We Know It. (October 18, 2024). Retrieved February 27, 2026.

<https://it.umd.edu/security-privacy-audit-risk-and-compliance-services-sparcs/topic-week/quantum-computing-how-it-changes-encryption-we-know-it>

² P.W. Shor. 1994. Algorithms for quantum computation: discrete logarithms and factoring. In *Proceedings 35th Annual Symposium on Foundations of Computer Science*, November 20-22 1994, Santa Fe, NM, USA, 124-134. <https://ieeexplore.ieee.org/document/365700/>

³ R. L. Rivest, A. Shamir, L. Adleman. A method for obtaining digital signatures and public-key cryptosystems. In *Communications of the ACM, Volume 21, Issue 2*, February 1, 1978. Retrieved April 24, 2026. Pages 120 - 126. <https://doi.org/10.1145/359340.359342>

one, to form a totient using the least common multiple of both $p-1$ and $q-1$ for the value of $\phi(n)$, which is kept hidden. The next step is to find another value (very often e , and will be defined so here as well for the purposes of demonstration) that is coprime with $\phi(n)$. That value is also part of the public key. Finally, search for a value that has congruency with $e^{-1}(\text{mod } \phi(n))$, which is then used as an exponent on the private key.

A quantum computer is necessary to efficiently run Shor's algorithm⁴, which was derived purely theoretically to take advantage of the nature of quantum characteristics, although there are large parts that still rely on classical computation. For instance, Shor's algorithm first attempts to reverse the steps involved in key generation of the RSA algorithm, effectively trying to find some value, a , with provable congruence to N such that $a \equiv 1(\text{mod } N)$. It does this to find values on the order of a . It is after this that Shor's algorithm uses Quantum Fourier Transforms for the purposes of finding the order r , where $a^r \equiv 1(\text{mod } N)$, which occurs with regular periodicity and can help determine the prime factors used to derive the key⁵.

Even though the concept of quantum decryption is presumed, the question remains about the mechanism of decryption. How can any computer execute enough calculations quickly enough to factor primes in a reasonable timeframe? Thanks to quantum bits, or qubits, calculations can be performed more quickly than with classical bits. A qubit can be defined in comparison to a classical bit via their states. "Just as a classical bit has a *state* - either a 0 or 1 - a qubit also has a state. Two possible states for a qubit are the states $|0\rangle$ and $|1\rangle$."⁶ Notation in the form of $\langle | \rangle$ is called bra-ket or Dirac notation, with the left facing notation being $\langle \text{bra} |$ and the right facing notation being $| \text{ket} \rangle$. A classical binary system sets the bits in two possible states. With the exception of voltage drift or perhaps an errant neutrino, the bits are absolute and can only refer to a single instance of information. Qubit states are exceptions to this restriction. Qubits can exist in altered states called superpositions, which are combinations of $|0\rangle$ and $|1\rangle$. The general form representing superpositions appears as:

⁴ Patrick Dreher. 2019. Shor's Algorithm. (November 19 & 21 2019). Retrieved February 17, 2026 from <https://moss.csc.ncsu.edu/~mueller/qc/qc19/readings/Shor%20Algorithm%20-%20Revised%20-%20typo%20fixed.pdf>

⁵ Dreher. Shor's Algorithm.

⁶ Michael A. Nielsen, Isaac L. Chuang, Massachusetts Institute of Technology. 2010. Quantum Bits. In *Quantum Computation and Quantum Information*. Simon Capelin, Margaret Patterson (Eds.) Cambridge University Press, 13

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle^7$$

The specific values of α and β aren't relevant as they only serve to represent the state of a qubit as wandering values across a 2D vector space, ultimately resulting in a value for $|\psi\rangle$ that is on the spectrum of 0 to 1. It behaves like the probability cloud of an electron around an atom's nucleus, which is so fundamentally obvious as to not need further explanation.

The process of decrypting RSA is a problem of factoring. Nobody has yet demonstrated that factoring on a classical computer can be achieved efficiently. A typical bit can hold $N = 2^n$ states, while the qubits can experience a broader spectrum of values and can efficiently calculate the Quantum Fourier Transform in $O(n \log n)$ time⁸. To return to the topic of period finding, Shor's algorithm attempts to factor a number, n . It can compute the function $a^r \equiv 1 \pmod{N}$ and prepare a super position over all values of r simultaneously using the qubits. Rather than processing potential values one by one and determining if they meet the criteria, all possibilities are evaluated in parallel, returning a result much sooner. The nature of a Quantum Fourier Transform causes constructive interference in cyclic patterns of $\frac{2^n}{r}$ and destructive interference with the same frequency, allowing for the period, r , to be accurately determined once again with classical post-processing, which can then be used to determine the original factors of the encryption method used.

As a quick side note, it is the nature of quantum mechanics to decohere^{9,10} with time. The specifics of which would overtake the theme of the paper, but the mere act of observing means that the result becomes less reliable over time and many more qubits are needed to perform calculations. The difficulty of acquiring qubits in close enough proximity to perform calculations is a limit and part of the reason why only an 80-bit

⁷ Nielsen and Chuang. *Quantum Computation and Quantum Information*. 13

⁸ L. Hales S. Hallgren. An improved quantum Fourier transform algorithm and applications. In *Proceedings 41st Annual Symposium on Foundations of Computer Science*, November 12-14 2000, Redondo Beach, CA, USA, 515-525. <https://ieeexplore.ieee.org/document/892139>

⁹ Quantum decoherence. From *Wikipedia*. Accessed 2026-03-02.

https://en.wikipedia.org/wiki/Quantum_decoherence

¹⁰ Craig Gidney, Martin Ekerå. *How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits*. April 13, 2021. <https://arxiv.org/abs/1905.09749>

encryption has been cracked, which is a far cry from the minimal 2048-bit RSA keys we see today.

With the advent of Shor's algorithm, it was known that there would eventually be a way to reliably crack RSA or Diffie-Hellman type encryption. The computer science community knew that while it was technologically infeasible for an immediate threat, there was still a degree of inevitability of the technology one day coming to fruition. With the goal of developing quantum resistant encryption algorithms, they developed new standardized methods for key-encapsulation and digital signatures¹¹.

However, there is a risk in over-relying on two of the three standard algorithms being based on the module-lattice principles. There may be a discovery of a flaw common to the underlying mathematics which can then be exploited by attackers. As it stands two of those algorithms cover digital signatures, yet only one covers key-encapsulation, making that lone area of encryption vulnerable to a state of total insecurity if broken. Regardless of the likelihood of the mathematics being broken in a timeframe that would affect the overall security of communication using the protocols, it is a near certainty that they will be broken eventually. RSA became the first publicly available commercial solution for public key exchange after it was described in 1977 and formally published in 1978¹², and now nearly 50 years on it has not been adequately broken to a point it is unusable. Even though regular communications use much higher bit encryptions than what has been demonstrably broken, it is not inconceivable that an arbitrary length encryption can be broken at some point in the future. It is better to move to a more secure standard while the capacity to easily decrypt RSA is still years or decades away.

The next step is introducing module lattice cryptography. A module is a mathematical structure that involves vectors, but over a ring instead of a field. The general module lattice theory uses algebraic methods to define a graph of nodes connected with a very particular relationship to each other. That relationship with ML-cryptography is fundamental, but very different in the ways that actually affect the

¹¹ 2024. NIST Releases First 3 Finalized Post-Quantum Encryption Standards. (August 13 2024). Retrieved February 17, 2026 from <https://www.nist.gov/news-events/news/2024/08/nist-releases-first-3-finalized-post-quantum-encryption-standards>

¹² Rivest, Shamir, Adleman. A method for obtaining digital signatures and public-key cryptosystems.

process of encoding a message. Without going too deep into the math¹³ of module lattices, module lattice frameworks can successfully emulate the prototypical Diffie-Hellman key exchange:

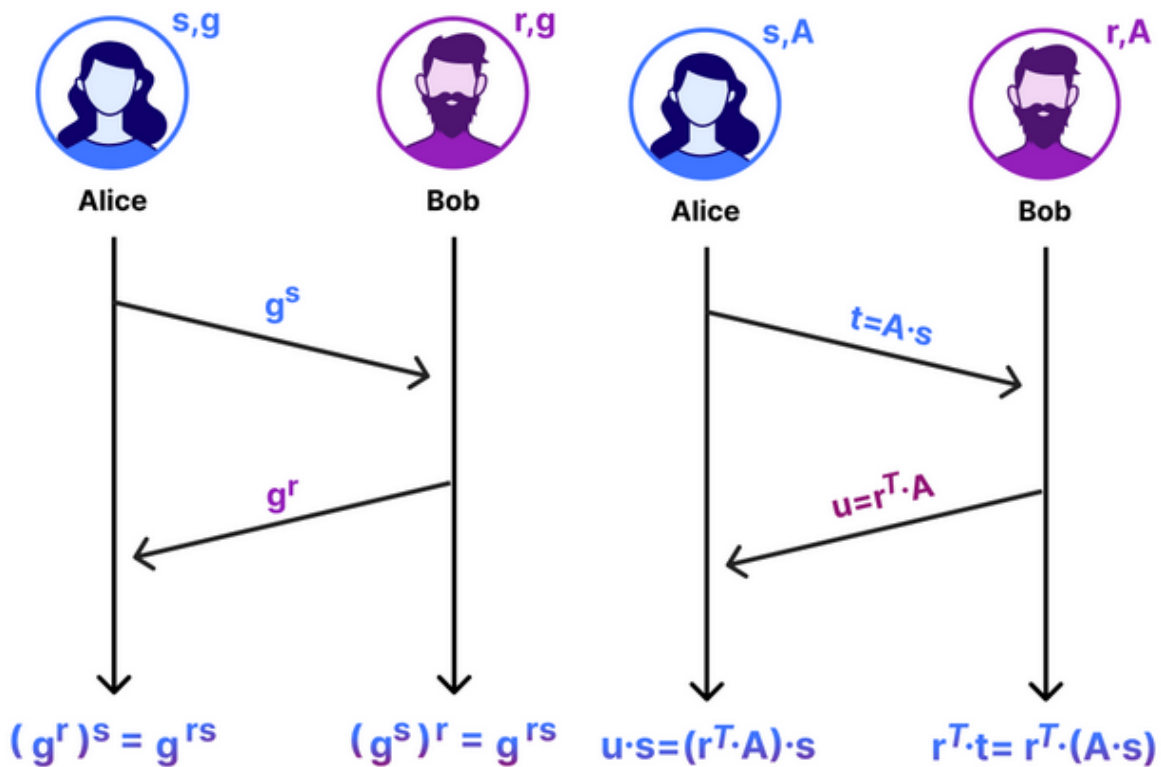


Figure 1: Classical key exchange (left) and ML-key encapsulation exchange (right). <https://blog.cloudflare.com/lattice-crypto-primer/>

Figure 1 shows a typical key exchange between Alice and Bob. They each use a random value to exchange between them and compute it in a way to confirm they both have the same shared value. ML-key exchange uses a similar approach, but with the calculation of a transverse matrix to compute vectors in a multidimensional vector space. On its own, that mere exchange is still vulnerable, but it can be made reasonably secure to mitigate attacks for the purpose of its regular use. However, this does present an area for attackers to focus on for a reducible attack surface.

¹³ Christopher Patton and Peter Schwabe. 2025. Prepping for post-quantum: a beginner's guide to lattice cryptography. The Cloudflare blog. Retrieved March 16, 2026 from <https://blog.cloudflare.com/lattice-crypto-primer/#all-the-math-we-need>

The interactions between values in a matrix can only be reduced to “less exponential”¹⁴ with current quantum algorithms, compared to the reduction to polynomial time for Shor’s algorithm attacking RSA. There are a number of hard problems¹⁵ to solve with cryptographic lattices. The complexity of modular arithmetic has the benefit of not being much easier to break using a quantum computer:

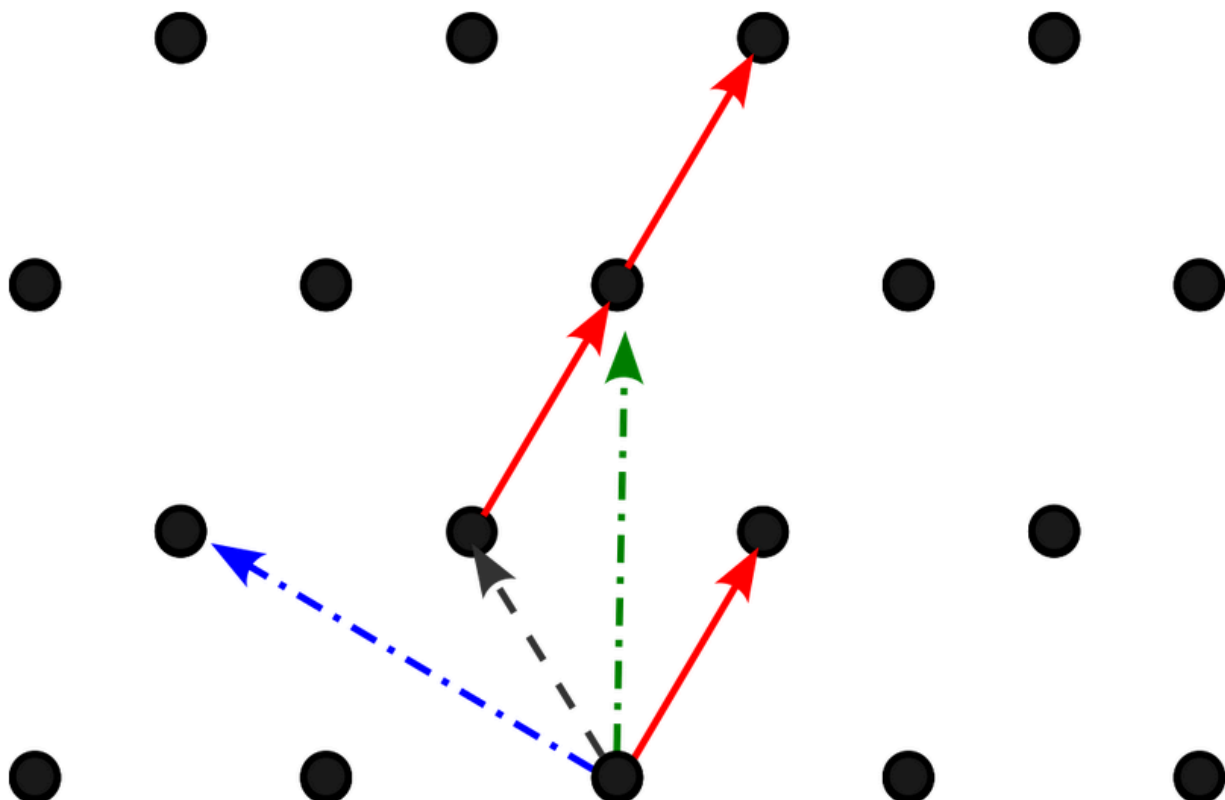


Figure 2: Example lattice. Multiple components can be used to create a vector.

<https://www.redhat.com/en/blog/post-quantum-cryptography-lattice-based-cryptography>

As pointed out in the Cloudflare blog, attacks such as Gaussian elimination can greatly reduce the size of the matrix¹⁶ used and provide a solvable problem for attackers. Even the ML methods mentioned so far required a number of other steps to obscure the weaknesses, and though they do reliably secure the algorithm for the time

¹⁴ Patton Schwabe. Prepping for post-quantum: a beginner’s guide to lattice cryptography.

¹⁵ Robert Relyea. 2025. Post-quantum cryptography: Lattice-based cryptography. The Redhat blog. Retrieved March 16, 2026 from

<https://www.redhat.com/en/blog/post-quantum-cryptography-lattice-based-cryptography>

¹⁶ Relyea. Post-quantum cryptography: Lattice-based cryptography.

being, it is not inconceivable for that fact to be the thin end of the wedge used to break open these algorithms.

The NIST paper¹⁷ introducing the new quantum-resistant standards even says, “The security of ML-KEM is based on the presumed hardness of the so-called Module Learning with Errors (MLWE) problem” and “The hardness of the MLWE problem is itself based on the presumed hardness of certain computational problems in module lattices”. The presumed hardness of the problem has not been proven to hold as of yet, “The main open question raised by this work is whether our results can be dequantized...the difficulty is that there seems to be no classical way to use an oracle that solves the closest vector problem within small distances.”¹⁸ Part of MLWE’s desirability is its ability to solve the shortest vector problem.

There may have been proofs since that guarantee the hardness of the MLWE problem, so there is some degree of security once the key encapsulation method is started. At the lowest level, given the inherent security flaws, more needs to be done in order to guarantee downstream protections. In CRYSTALS-Kyber, the version that predates standard for quantum-resistant KEM, a Fujisaki-Okamoto transform¹⁹ is applied to a Kyber-generated key in an effort to protect a message from both chosen-plaintext attacks (CPA-secure) and chosen-ciphertext attacks (CCA-secure). Once the key is encapsulated in a manner that is CCA-Secure, then the key can be encrypted, exchanged, and authenticated using the Kyber encapsulation and decapsulation algorithms.

It is worth mentioning is the fact that ML-KEM uses the number-theoretic transform to calculate polynomials on the ring and that allows for some flexibility while maintaining a higher rate of calculation than comparable algorithms²⁰. Once

¹⁷ 2024. Module-Lattice-Based Key-Encapsulation Mechanism Standard. (August 13, 2024). Retrieved February 17, 2026 from <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.203.pdf>

¹⁸ Regev, O. On lattices, learning with errors, random linear codes, and cryptography. In Proceedings of the 37th ACM Symposium on Theory of Computing, Baltimore, MD, USA, 22–24 May 2005; pp. 84–93. <https://dl.acm.org/doi/epdf/10.1145/1060590.1060603>

¹⁹ Eiichiro Fujisaki and Tatsuaki Okamoto. Secure integration of asymmetric and symmetric encryption schemes. In Michael J. Wiener, editor, CRYPTO’99, volume 1666 of LNCS, pages 537–554. Springer, Heidelberg, August 1999. <https://dl.acm.org/doi/10.5555/646764.706343>

²⁰ J. Bos, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, J. M. Schanck, P. Schwabe, G. Seiler, and D. Stehle, “CRYSTALS–kyber: A CCA-secure module-lattice-based KEM,” in *Proc. IEEE Eur. Symp. Secur. Privacy (EuroS&P)*, Apr. 2018, pp. 353–367. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8406610>

transformed, a ring R_q has a companion in the form of T_q . This allows a user to to change the value of the rank-'k' over the ring, allowing a single codebase for three security levels of Kyber-512/768/1024²¹.

$$f \times_{R_q} g = \text{NTT}^{-1}(\hat{f} \times_{T_q} \hat{g}),$$

Figure 3. Ring R_q is the computational inverse of ring T_q .
<https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.203.pdf>

ML-KEM incorporates quantum-resistant mechanisms into the Kyber algorithms, outputting 256-bit values for the encaps and decaps algorithms securely enough that there is an overwhelming chance of success of securing the key using the ML-KEM

methods. The choice of the ring $R_q = \frac{\mathbb{Z}_q[X]}{(X^{256} + 1)}$ allows a cyclotomic that allows a Number Theoretic Transform (NTT) with $q = 3329$ chosen as the smallest prime that makes $q \equiv 1 \pmod{512}$ work. They had the foresight to implement the destruction of intermediary data in the algorithms, reducing the likelihood of a successful exfil of data partway through an encryption process.

The mathematics behind the Module-Lattice-based Digital Signature Algorithm (ML-DSA) are quite similar. The basis of ML-DSA, CRYSTALS-Dilithium²², still generates a key with a ring of polynomials, and it then signs and verifies data passed through, using other algorithms to randomize it and perform hash functions to disperse the results for further verification. The upshot is meant to provide total security, “ML-DSA is designed to be strongly existentially unforgeable under chosen message attack (SUF-CMA). That is, it is expected that even if an adversary can get the honest party to sign arbitrary messages, the adversary cannot create any additional valid signatures based on the signer’s public key, including on messages for which the signer has already provided a signature”²³.

²¹ Bos, et al. CRYSTALS-kyber: A CCA-secure module-lattice-based KEM. 354

²² S. Bai, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, P. Schwabe, G. Seiler, and D. Stehle, “CRYSTALS-Dilithium Algorithm Specifications and Supporting Documentation (Version 3.1)”. Accessed 2026-03-17 <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8406610>

²³ 2024. Module-Lattice-Based Digital Signature Standard. (August 13, 2024). Retrieved February 17, 2026 from <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.204.pdf>

ML-DSA consists of 3 steps: Commitment, Challenge, and Response. The prover sends a value to the verifier who sends another value to the prover, who must return a value to the verifier who will then check to confirm all the values match an expected return value. This reduces the likelihood of a forged certificate to a negligible degree. The commitment value the signer uses, y , which is similar to a nonce in other algorithms like AES. Reusing the commitment value does open up the algorithm to a pattern analysis attack. Thus, the randomness of that value is crucial in order to guarantee security. The underlying Dilithium algorithm rejects samples based on the response based on a Fiat-Shamir with aborts²⁴ process. If signing would leak any information about the secret, the process can be halted, thrown away, and tried again.

The necessity of digital signatures can help verify proof of possession and the identity of a party that is in communication with the user. The necessity to verify the identity is great enough to warrant a secondary DSA method, the Stateless Hash-Based Digital Signature Standard²⁵ (SLH-DSA) has also been set as a standard. That statelessness is a major feature of security, as it won't have the opportunity to reuse one-time keys, effectively removing pattern analysis attacks as a potential attack vector. SPHINCS⁺²⁶ is the basis of the current standard. In addition to being stateless, it is an extended Merkle tree (hypertree), using many few-time signatures (FTS) key pairs in the tree, which itself consists of hash-based many-time signatures (MTS), the top level of which signs the MTS public keys below it, which sign the keys below that, and so on, with the second to last layer of MTS public keys signing the FTS key pairs and uses WOTS⁺ for encryption. The cost of using this architecture is that it is slower than ML-DSA, even though the verification is faster.

SLH-DSA improves on this by using pseudorandom functions for a Forest of Random Subsets (FORS) and WOTS²⁷, separating the domain of the prf calls from the

²⁴ Vadim Lyubashevsky. Fiat-Shamir with Aborts: Applications to Lattice and Factoring-Based Signatures. 2009. *ASIACRYPT '09: Proceedings of the 15th International Conference on the Theory and Application of Cryptology and Information Security: Advances in Cryptology* Pages 598 - 616.

https://doi.org/10.1007/978-3-642-10366-7_35

²⁵ 2024. Stateless Hash-Based Digital Signature Standard. (August 13, 2024). Retrieved February 17, 2026 from <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.205.pdf>

²⁶ SPHINCS+. Stateless hash-based signatures. Retrieved from <https://sphincs.org/>

²⁷ Stateless Hash-Based Digital Signature Standard.

rest of the routine. They follow the same domain separation for the private key using a PK.seed. The top of the tree is the PK.root, which is a public key of the XMSS key. The overall structure appears as so:

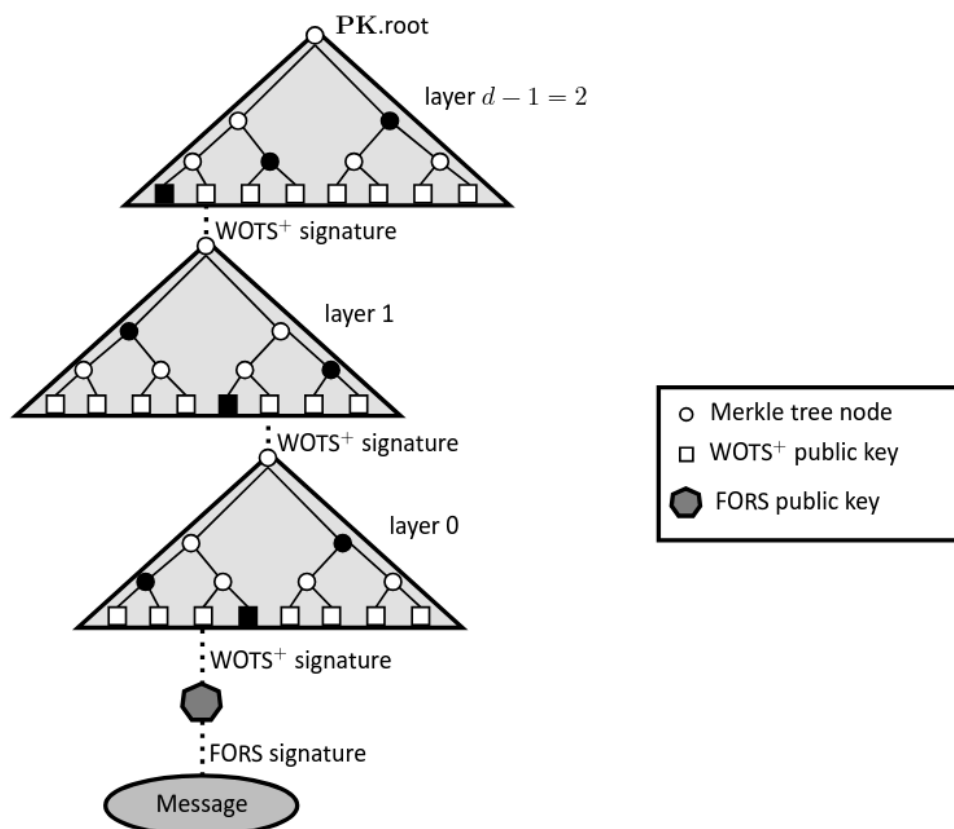


Figure 4. SLH-DSA Signature structure

<https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.205.pdf>

In the SLH-DSA, there is no underlying mathematical structure to exploit. The current structure of the hypertree requires an extensive search. The current best quantum algorithm is Grover's algorithm²⁸, which runs an unstructured search and provides a quadratic speedup over typical tree traversal methods. Grover's algorithm also doesn't parallelize well, reducing the effectiveness of other processors assisting the

²⁸ Lov K. Grover. A fast quantum mechanical algorithm for database search From *STOC '96: Proceedings of the twenty-eighth annual ACM symposium on Theory of Computing*. Pages 212 - 219. <https://doi.org/10.1145/237814.23786>

search. Shor's algorithm is able to use Fourier transforms applied to the algebraic structure to reduce the calculation cost exponentially.

All of that information is presented not to deride the quantum-resistant standards or call them insufficient. Indeed, teams of highly educated people came up with a number of clever methods to encrypt data with a reasonable degree of certainty in the security of the system. As mentioned earlier in the paper, RSA has been the most widely available tool for 50 years to encrypt messages without sandbagging transmission or practical use. Until recently, most cases where it failed didn't involve cracking the keys, but other methods of stealing or co-opting those keys. The new standards met several requirements for security, with only reasonable trade-offs for speed. If there is a better process for determining a type of encryption that is verifiably secure from current and known future threats, it hasn't been dreamt of by the cryptographic community as of yet.

In WWII, the Germans used the Enigma machine as a method of encrypting their communications. It was a clever device, but Polish cryptographers managed to crack it before the war, and the supplied information gave British SIGINT enough information to finally crack codes even after improvements to the device were made. Germany's presumption of security led to many of their early losses. We should not presume that a single method of encryption is secure, either.

To that point, a wider variety of tools that can be quickly distributed in the instance of total insecurity should be in place. Two of the three current standard algorithms that meet quantum resistance rely on Module Lattice based mathematics. If a weakness is found in either of those methods, then there is no standard that can be used to resist quantum attacks. It must be noted that the first known algorithm with the capability of breaking RSA in a reasonable time came out 17 years after RSA's introduction. RSA-1024²⁹ was deprecated 2 years ago because it can theoretically be broken by a well-funded actor using classical computational methods, resulting in the cryptographic community making RSA-2048 the current standard. Even if classical computational power improved drastically by following Moore's law and reducing the

²⁹ Elaine Barker, Allen Roginsky. *Transitioning the Use of Cryptographic Algorithms and Key Lengths*. March 2019. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-131Ar2.pdf>

compute time by half every few years, it would also hypothetically be easier to increase the number of bits in the RSA standard to compensate. Even so, the increasing quality and power of quantum computers is bringing the number of secure bits in RSA down. For that reason, the settings of these new standards is a terrific first step in the continuing security of data communications. There were other finalists in the quest to find quantum resistant algorithms. The community can once again come together to shore up their weaknesses and make them ready for deployment as a contingency. It may even be wise for private organizations with the resources to deploy those systems privately, as there isn't as much of an appeal to focus on them as targets to attempt to weaken.

The reason for doing so is borne out by history. There have been multiple instances in the past of standards being broken. Data Encryption Standard (DES) was weak from its introduction, with only an effective 56-bit key size. It was effectively broken in January 1999 in less than a day's worth of computing power³⁰. Any motivated attacker could break that with little effort. There was also controversy surrounding the involvement of the NSA in the creation of the standard with suspicions they created a back door. Regardless, DES is still in use today in the form of Triple DES in legacy applications despite NIST withdrawing the FIPS document supporting it³¹. DES may be deprecated, but it's still an active security risk.

SHA-1³² is another case of an effectively broken former standard of a hash instead of direct encryption. With the SHattered attack presented by Google and CWI Amsterdam, industry leaders moved to remove the most visible use of SHA-1, TLS, making the deprecation more extensive than the reduction of DES. Further attacks since the introduction of SHattered have reduced the time required to forge a certificate. This too has stronger versions available that cover the holes that were present in that version. SHA-1 itself is an improved version of what is now called SHA-0, which had its own defects requiring rapid deprecation and replacement. When sufficiently motivated

³⁰RSA Code-Breaking Contest Again Won by Distributed.Net and Electronic Frontier Foundation (EFF). From eff.org. Accessed 2026-04-24.

https://w2.eff.org/Privacy/Crypto/Crypto_misc/DESCracker/HTML/19990119_deschallenge3.html

³¹ Barker and Roginsky. *Transitioning the Use of Cryptographic Algorithms and Key Lengths*.

³² Marc Stevens, Elie Bursztein, Pierre Karpman, Ange Albertini, Yarik Markov. *The first collision for full SHA-1*. July, 29 2017. https://link.springer.com/chapter/10.1007/978-3-319-63688-7_19

the community can in fact replace insecure algorithms. While acknowledging that being able to quickly remove encryption processes isn't 'safe', the less friction introduced for the replacement of insecure algorithms, the more security we can provide.

The last insecure algorithm to discuss is the dual EC DRBG³³. Originally published as a standard in 2006, it had a relatively short life cycle after being withdrawn in 2014. Many members of the cryptographic security community raised concerns about the security of dual EC DRBG. In fact, it seems the NSA had its very own backdoor with this particular algorithm. The concerns about the DES including a backdoor may not have been unfounded at all.

Finally, another algorithm with a lot of promise was Supersingular isogeny Diffie-Hellman (SIDH)³⁴. It was able to adequately resist attacks with a relatively small key size, making it fast and efficient. In the instance that a key was ever recovered, it couldn't necessarily be used to decrypt old data. However, in 2022, a key-recovery attack was able to successfully extract keys with very little classical computing power. The structure of the problem perhaps could have been exceptionally resilient against attacks made by a quantum computer, but failed against comparatively simple algebraic computation.

The current standards are not a cause for concern. It is the overreliance on a single type of standard that can lead to an unacceptable level of vulnerability for any entity actively using it. RSA has not proven to have any structural vulnerabilities since its inception into regular use, and that very well may be the case for the module lattice family of algorithms is at least as sustainable as RSA and has an equivalent lifetime before being phased out. The inertia of systems and organizations that can't automatically be updated due to internal logistics or external regulations mean that a potentially insecure system can incur risks well after the exposure of a weakness. It is in the best interest of the community to continue to develop other cryptographic methods until they meet the same security requirements as the standards.

³³ D.J. Bernstein, T. Lange, R.F. Niederhagen. *Dual EC : a standardized back door*. International Association for Cryptologic Research. 2015.

<https://research.tue.nl/en/publications/dual-ec-a-standardized-back-door/>

³⁴ Castryck, W., Decru, T. *An Efficient Key Recovery Attack on SIDH*. In: Hazay, C., Stam, M. (eds) *Advances in Cryptology – EUROCRYPT 2023*. EUROCRYPT 2023. Lecture Notes in Computer Science, vol 14008. Springer, Cham. https://doi.org/10.1007/978-3-031-30589-4_15

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